



LINUX Interview Style Q&A with Practical Context DAY 12

Networking

51.You're troubleshooting a server and need to know its IP address. How would you find it?

A: I'd use `ip addr` to display all network interfaces and their IP addresses. It's the modern replacement for `ifconfig`, which is deprecated on many systems.

Example:

```
ip addr show
```

52.How do you check which network ports are listening on your system?

A: I'd use `ss -tuln`, which shows active TCP and UDP listening sockets without resolving hostnames.

Alternative (older): `netstat -tuln`

Example:

```
ss -tuln
```

53.How do you check if a remote host is reachable from your machine?

A: I'd use the `ping` command:

```
ping google.com
```

It sends ICMP packets and waits for replies to test basic connectivity.

54.A remote service runs on port 8080. How do you test if that port is accessible?

A: I'd use `telnet` like this:

```
telnet remote_host 8080
```

If `telnet` is unavailable, I could use `nc -zv remote_host 8080` to test TCP port connectivity.

55.How do you view the routing table of a Linux system?

A: I'd use the modern `ip route` command to display current routes.

Example:

```
ip route
```

56.You're trying to find which process is using a specific port. How do you do that?

A: I'd run:

```
lsof -i :<port_number>
```

This lists all processes using the specified port. For a full list of open network connections, just use `lsof -i`.

57.How do you download a file directly from the command line?

A: I'd use `wget` like so:

```
wget https://example.com/file.tar.gz
```

It's a non-interactive downloader—ideal for automation or remote installations.

58.You need to copy a file from your local machine to a remote server securely. What command would you use?

A: I'd use `scp`:

```
scp myfile.txt user@remote_host:/path/to/destination/
```

This securely copies files over SSH.

59.How do you remotely access a Linux server from your terminal?

A: I'd use ssh:

```
ssh user@remote_host
```

If needed, I can specify a port with -p or a key with -i /path/to/key.pem.

60.How would you assign a static IP address to a Linux system?

A: It depends on the distribution and the networking backend:

- Debian/Ubuntu (legacy):
Edit /etc/network/interfaces
- Red Hat/CentOS:
Edit /etc/sysconfig/network-scripts/ifcfg-eth0
- Ubuntu (Netplan-based):
Edit YAML files in /etc/netplan/ and run netplan apply

Example for Netplan:

yaml

network:

version: 2

ethernets:

eth0:

addresses: [192.168.1.100/24]

gateway4: 192.168.1.1

nameservers:

addresses: [8.8.8.8, 1.1.1.1]

User Management

61.How do you create a new user in Linux?

A: You can use the adduser command (user-friendly) or useradd (lower-level).

```
adduser alice
```

This creates a user, home directory, and prompts for a password.

62.How do you delete a user from the system?

A:To delete a user and optionally their home directory:

```
userdel alice      # removes user only
```

```
userdel -r alice   # removes user and home directory
```

63.How do you lock a user account temporarily?

A:Use the passwd -l (lock) command:

```
passwd -l alice
```

This disables login without deleting the user.

64.How do you unlock a previously locked user account?

A:Use the passwd -u (unlock) command:

```
passwd -u alice
```

65.How do you switch to another user from the terminal?

A:Use the su (substitute user) command:

```
su - alice
```

The - ensures the user's environment and login shell are loaded.

66.How do you see who is currently logged into the system?

A: Use either of these:

`who` # shows login terminals

`w` # shows more detail (time, processes)

67.How do you display information about a specific user?

A: Use the `id` command:

`id alice`

Shows UID, GID, and group memberships.

68.How can you list all the users on a Linux system?

A: Check the `/etc/passwd` file:

`cat /etc/passwd`

Each line represents a user. You can filter real users with:

`cut -d: -f1 /etc/passwd`

69.How do you create a new group?

A: Use the `groupadd` command:

`groupadd developers`

70.How do you add a user to a specific group?

A: Use `usermod` with the `-aG` option:

`usermod -aG developers alice`

`-a` appends to group, `-G` specifies the group. Omitting `-a` will overwrite existing groups.

71.How do you change a user's default shell?

A: Use the `chsh` command:

`chsh -s /bin/bash alice`

72.How do you change the home directory of a user?

A: Use `usermod` with `-d` and `-m` to move existing files:

`usermod -d /home/new_alice -m alice`

73.How do you create a system user without a home directory?

A: Use:

`adduser --system --no-create-home nginx`

System users are typically used for running daemons or services.

74.How do you set or reset a user's password?

A:Use the `passwd` command:

`passwd alice`

It prompts for a new password.

75.How do you check when a user account's password was last changed?

A: Use the `chage` command:

`chage -l alice`

This shows password aging info, last change, expiry, etc.

Package Management

76.How do you install a package on a Debian-based system like Ubuntu?

A:You use the apt command:

```
sudo apt install package_name
```

Example:

```
sudo apt install nginx
```

This command downloads and installs the package along with any required dependencies.

77.How do you update all packages to the latest version on Debian-based systems?

A:First, update the package index, then upgrade installed packages:

```
sudo apt update && sudo apt upgrade
```

- apt update: fetches latest package list from repositories.
- apt upgrade: installs the latest versions of installed packages.

78.How do you remove a package but keep its configuration files on Debian systems?

A:Use the apt remove command:

```
sudo apt remove package_name
```

To remove everything including configs:

```
sudo apt purge package_name
```

79.How do you install a package on RHEL-based systems (Red Hat, CentOS, Fedora)?

A:Use either yum or dnf (newer tool):

```
sudo yum install package_name
```

or

```
sudo dnf install package_name
```

Example:

```
sudo dnf install httpd
```

80.How do you remove a package on RHEL-based systems?

A:Use:

```
sudo yum remove package_name
```

or

```
sudo dnf remove package_name
```

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or

```
sudo dnf remove package_name
```

81.How do you search for a package in Debian-based systems?

A:Use: apt search package_name

This helps find the exact package name if you're unsure.

82.How do you list all installed packages on a system?

A:

- On Debian-based systems:

```
dpkg -l
```

- On RHEL-based systems:

```
rpm -qa
```

You can also pipe through grep to search:

```
rpm -qa | grep nginx
```

83.How do you check which package a file belongs to?

A:

Debian: `dpkg -S /path/to/file`

RHEL: `rpm -qf /path/to/file`

Useful for tracking file ownership when troubleshooting.

84.How do you clean up cached .deb package files on Debian-based systems?

A:Use:

`sudo apt clean`

This clears the local cache located at `/var/cache/apt/archives`.

85.How do you enable or disable a repository in RHEL-based systems?

A:

Use `yum-config-manager` (part of `yum-utils`):

`sudo yum-config-manager --enable repo_name`

`sudo yum-config-manager --disable repo_name`

Example:

`sudo yum-config-manager --disable epel`

System Monitoring and Performance

86.How do you check how long the system has been running?

A:Use the `uptime` command:

`uptime`

Displays current time, uptime duration, number of users, and system load averages.

87.How do you display memory usage on the system?

A:Use:

`free -h`

- `-h` gives human-readable format (MB/GB).
- Shows total, used, and available memory (RAM + swap).

88.How do you monitor disk I/O in real time?

A:Use the `iostat` command (from `sysstat` package):

`iostat -dx 2`

- `-d`: device stats
- `-x`: extended statistics
- `2`: refresh every 2 seconds
- Useful to identify slow disks or high I/O wait.

89.How can you check CPU usage live?

A:Use:

`top`

or better:

`htop`

`htop` is more interactive and color-coded, making it easier to analyze CPU/memory consumption by process.

90.How do you list currently running services?

A:Use systemctl:

```
systemctl list-units --type=service
```

Displays active, loaded, or failed system services managed by systemd.

91.How do you enable a service to start automatically at boot time?

A:Use:

```
sudo systemctl enable nginx
```

Replaces older chkconfig used in SysVinit systems.

92.How do you view system load averages?

A:Either:

```
uptime
```

or:

```
cat /proc/loadavg
```

Shows 1, 5, and 15-minute load averages.

A value > number of CPU cores means the system is under high load.

93.How do you monitor network bandwidth in real time?

A:Use tools like:

```
iftop
```

or:

```
nload
```

They show live incoming/outgoing traffic per interface. May need installation.

94.How do you list detailed hardware information?

A:Use:

```
sudo lshw
```

Provides comprehensive hardware info—CPU, RAM, storage, network, etc.

Note: Requires root privileges.

95.How do you check the current kernel version?

A:Use:

```
uname -r
```

Outputs the running Linux kernel version (e.g., 5.15.0-84-generic).

96.How do you view all currently mounted filesystems?

A:Use:

```
mount
```

or:

```
df -h
```

- df -h shows disk space usage in human-readable format per mount point.

97.How do you list the current user's scheduled cron jobs?

A:Use:

```
crontab -l
```

Lists cron jobs in the current user's crontab.

98.How do you check system logs in real-time?

A:Use:

`tail -f /var/log/syslog` # For Ubuntu/Debian

`journalctl -xe` # For systemd-based logs

`tail -f` is useful for real-time log monitoring.

99.How do you analyze disk usage by directory size?

A:Use:

`du -h --max-depth=1 /path`

- Shows sizes of subdirectories under /path.
- `--max-depth=1`: limits depth to top-level directories.
- `-h`: human-readable sizes (MB, GB).